

Literature review protocol — draft v0.1

Project: Multimodal RAG for Enterprise Email and Attachments (PFE 3358)

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Everything below is a proposal. Points marked **[decide]** are collected in section 8.

1. Review type and reporting standard

Proposed: a **scoping review**, reported with PRISMA-ScR (Tricco et al., 2018), with the search reported as PRISMA-S asks and the PRISMA 2020 flow diagram in its "databases, registers and other sources" version.

[decide]

Reason: the project spans four sub-areas with different kinds of studies, and the aim is to map what exists and document a gap, not to pool results. The PRISMA 2020 items on risk of bias, effect measures and certainty of evidence do not apply and will be marked as such.

2. Review questions

	Question	Thesis RQ
LQ1	Which datasets and benchmarks exist for RAG over email or other enterprise data, and what do they annotate (threads, metadata, attachments, duplicates, conflicts)?	RQ4
LQ2	How do RAG systems represent the links between a message, its attachments, its thread and its metadata?	RQ2
LQ3	How are duplicates, redundancy and knowledge conflicts, especially temporal ones, detected, resolved and evaluated in RAG?	RQ1
LQ4	How are images and visually rich documents integrated into RAG (multimodal embedding, OCR, captioning, fusion), and how do the options compare?	RQ3

3. Eligibility criteria

	Include	Exclude
Period	Published or posted from 2020-01-01, the year RAG was introduced	Earlier work, except as background
Language	English	Other languages
Type	Peer-reviewed papers and arXiv preprints [decide]	Abstracts, posters, tutorials, papers under 4 pages
Topic	Addresses at least one of LQ1–LQ4	Email classification (spam, phishing, routing) without retrieval-augmented generation; security or privacy attacks on RAG where email is only the example
Evidence	Presents a system, dataset, benchmark or empirical evaluation	No evaluation
Versions	The published version when one exists	The preprint of a paper that was later published

Surveys are kept for citation searching and are not charted.

Why preprints are proposed: in the arXiv run of S1, 228 of 270 records were submitted in 2025 or 2026, and 9 of the proposal's 24 references are arXiv-only.

4. Information sources

Source	Why	Access	Status
arXiv	Preprints; most of this field appears here first	Open API	Run on 2026-10-06
Scopus	Broad index of peer-reviewed venues	Institutional [decide]	To run
IEEE Xplore	IEEE conferences and journals	Search is open, by hand only	Not run; reached through OpenAlex
ACM Digital Library	SIGIR, CIKM, KDD and other ACM venues	Search is open, by hand only	Not run; reached through OpenAlex
OpenAlex	Open index of scholarly works, including IEEE and ACM publications	Open API	Run on 2026-10-06 for IEEE- and ACM-published papers [decide]
ACL Anthology	ACL, EMNLP, NAACL, COLING	Open, but no Boolean search	Run on 2026-10-06 by script; method to confirm [decide]
Other methods	The 24 references of the proposal; backward and forward citation searching from included papers	—	References loaded

For the ACL Anthology, the proposal is to filter its full bibliography file (`anthology+abstracts.bib.gz`, which includes abstracts) with the same concept blocks by script, so the search is reproducible. A first run on 2026-10-06 gave 273 unique records, 174 of which are also in the arXiv results.

IEEE Xplore and the ACM Digital Library cannot be queried by script. A first run on 2026-10-06 reached their publications through OpenAlex instead, by DOI prefix: 1,034 unique records, 190 of which are also in the arXiv results. OpenAlex has no abstract for 64 of them, so its recall is lower than a search in the two databases themselves.

5. Search strategy

Concept blocks, searched in title and abstract (and keywords where the database has them):

Block	Terms
RAG	"retrieval augmented" OR RAG
CONTEXT	email* OR e-mail* OR mailbox OR inbox OR enterprise OR "company-internal" OR workplace
EMAIL	email* OR e-mail*
HYGIENE	"knowledge conflict*" OR "conflicting evidence" OR deduplication OR "near-duplicate" OR redundancy OR outdated OR superseded
MODALITY	multimodal OR "multi-modal" OR "visual document" OR OCR
ATTACHMENT	attachment* OR multimodal OR "multi-modal"

ID	String	Serves
S1	RAG AND CONTEXT	LQ1, LQ2
S2	RAG AND HYGIENE	LQ3
S3	RAG AND MODALITY	LQ4
S0	RAG AND EMAIL AND ATTACHMENT	Gap check: expected to be nearly empty

The strings are adapted to each database's syntax, and each adapted string is written into `search-log.md` exactly as run, with the date, limits and hit count.

Validation against known papers. 21 of the proposal's 24 references are indexed on arXiv. The strings retrieve 14 of them. The 7 they miss are background or method papers:

- [1] RAG, [6] Ragnarök and [15] RAGAS are general RAG works with no topic term.
- [7] training-data deduplication, [9] CLIP, [22] ConflictBank and [23] fusion functions are not RAG papers.

These seven enter through "other methods". Whether LQ3 needs a second string without the RAG block, to catch work like ConflictBank, is open. **[decide]**

6. Selection process

1. All records go into the Zotero group library FrAInderl-RAG-Review. Duplicates are removed and the number removed is logged.
2. Both reviewers screen titles and abstracts independently against section 3. A pilot on 50 records, drawn at random across S1 to S3, calibrates the criteria first. Agreement is reported as Cohen's kappa. Disagreements are settled by discussion, then by the supervisor.
3. Full texts of the remaining reports are assessed, and one exclusion reason is recorded for each report excluded.

Tool for blind dual screening: Rayyan, or separate tag prefixes per reviewer in Zotero. **[decide]**

7. Data charting

One row per included paper: reference, year and venue; review question(s) addressed; data type (email, enterprise, documents); modalities; dataset or benchmark used or introduced; method; metrics; main result; limitations; relevance to the thesis RQs.

8. Open decisions for 2026-10-08

1. Scoping review (PRISMA-ScR) or full systematic review?
2. Are arXiv preprints eligible?
3. Is the 2020 start date acceptable?
4. Which subscription databases can THD give access to (Scopus, Web of Science)?
5. How should the ACL Anthology be searched?
6. Is OpenAlex acceptable for reaching IEEE and ACM publications, or must IEEE Xplore and the ACM Digital Library be searched directly?
7. S3 accounts for 1,573 of the 2,555 unique records so far. Narrow it, or keep it broad although RQ3 is a stretch goal?
8. Does LQ3 need a string without the RAG block?
9. Which screening tool, and who arbitrates disagreements?